Chemical reachability limits mutagen-based TKI-resistance screens

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Abstract

Background. Preclinical Ba/F3 mutagenesis screens are a first line of evidence for predicting kinase-inhibitor resistance, and their hit lists shape resistance panels for KRAS G12C, ALK, and MET inhibitors. Every mutagen in use has a non-uniform substitution spectrum, and whether that chemistry limits which resistance variants a screen can find, or only reorders them, has not been audited against a chemistry-free baseline.

Methods. We separate a screen's expected output into a supply term, from codon tables and each mutagen's six-class spectrum, and a selection term, from chemistry-free deep mutational scans of KRAS G12C, ALK, and MET: $P(v) \propto S_{\text{mutagen}}(v) \cdot \exp(\beta\sigma(v))$. We test supply by enumerating what each chemistry cannot reach, and selection by asking whether supply improves the ranking of the hits four retrospective screens reported.

Results. Chemistry acts as a reachability threshold, not a ranking weight. Base editors, which produce only two of the six substitution classes, cannot supply seven of the twelve KRAS G12C panel variants or most of each ALK drug's top-ranked variants. A second limit binds every single-nucleotide mutagen: roughly three-quarters of each MET inhibitor's top 100 priorities need two or more nucleotide changes. Past that threshold supply does not improve the ranking on any anchor, and mutagens with very different spectra recover ALK variants at indistinguishable rates. Two MET screens sharing one mutagen and one cell line gave different class compositions ($P = 0.029$), so composition follows the drug, though clone counts within a single codon did track supply.

Conclusions. The deliverable is the reachability inventory: which resistance variants a mutagen-and-drug pair cannot produce, enumerable from codon tables and a published spectrum before the screen runs. Reweighting the output afterward does not recover them, so chemistry belongs in a screen's design, not in its interpretation.

Keywords. Preclinical resistance screens; mutational signatures; ENU; Mutazyme; base editing; prime editing; deep mutational scanning; KRAS G12C; sotorasib; adagrasib; ALK; alectinib; lorlatinib; neladalkib; chemistry bias.

1 Background

Preclinical mutagenesis of Ba/F3 cells transduced with an oncogenic kinase is the standard first-line assay for predicting the resistance variants that will emerge in patients treated with a given tyrosine-kinase inhibitor (TKI) [1–4]. The assay was first developed for BCR-ABL1 and has since been applied to the resistance landscapes of ALK, FLT3, EGFR, KIT, FGFR, MET, ROS1, and KRAS G12C [2, 4–13]. Its appeal is that the output ranks variants by relative resistance in cells, and that the top-ranked variants have reproducibly matched the mutations that later appeared on ctDNA liquid-biopsy panels of relapsed patients [8, 9].

The assay rests on a mutagenesis step, and that step is almost always done with a chemistry-biased mutagen. In current (2024–2026) practice, an industry screen of the fourth-generation ALK TKI neladalkib (NVL-655) used ENU [14]; a next-generation ALK TKI panel screen covering alectinib, lorlatinib, zotizalkib, and neladalkib used error-prone PCR [7]; and a recent KRAS G12C resistance screen against adagrasib and the RAS(ON) tri-complex inhibitors used a combination of CRISPR cytosine and adenine base editors [15]. Each of these chemistries supplies the six pyrimidine substitution classes (C>A, C>G, C>T, T>A, T>C, T>G) at distinct probabilities: ENU is dominated by T>C and T>A, Mutazyme II by C>T, and base editing is restricted by design to C>T and T>C. The resulting six-class supply pattern is distinct for every chemistry. Because the hit list these screens produce is what informs resistance panels and treatment-sequencing decisions, a variant the chemistry cannot reach is not merely absent from an experiment. It is absent from the evidence base that clinicians and trial designers consult, and no amount of sequencing depth will recover it. We quantify two things. First, which reported resistance variants each chemistry cannot produce in a single nucleotide step, separating those observed in patients from those observed only in an accompanying *in vitro* scan. Second, for the variants it can produce, whether their ranking differs from the chemistry-free one.

Chemistry bias in mutagen output is not a new observation. The Mutazyme II spectrum is quantified in [2]'s supplement, ENU's class preferences are compiled in [16], the industry neladalkib poster itself states that ENU "has mutational bias" and that this is "a potential limitation of this assay" [14], and a Hamming-distance-1 reachability filter has been applied case-by-case to specific EGFR variants [17]. Rather than claiming a novel bias, this work systematizes that filter across mutagens, genes, and drugs, anchors it in chemistry-free atlases, reports it prospectively with per-variant codon-level provenance, and benchmarks it against both the σ-alone and the supply-alone nulls.

Two recent publications supply the missing baselines (Table 1, bottom two rows). A Twist Biosciences synthetic saturation mutagenesis library [10] produced every single-amino-acid substitution of KRAS G12C in a chemistry-free pool, and scored each variant under sotorasib

and adagrasib in NCI-H358 cells; in parallel, a prime-editing atlas [11] generated 3,208 single-nucleotide variants across the ALK kinase-domain locus (2,953 coding SNVs covering the kinase domain plus 255 flanking intronic and splice-site variants) in H3122 ALK+ non-small-cell lung cancer cells and scored each under alectinib, lorlatinib, and zotizalkib. In both studies, library construction is decoupled from mutagen chemistry, so the measured drug-selection score for each variant can be read as the fitness ground truth $\sigma(v)$ against which chemistry-biased screens can be compared. Several further chemistry-free atlases, independent of this work's two case studies, extend the template to other kinases. For *MET*, a kinase-domain DMS [18]: a Twist-synthesized library of 5,764 substitutions scored in Ba/F3 against eleven *MET* inhibitors spanning type I, I$\frac{1}{2}$, II, and III mechanistic classes. For *EGFR*, two atlases: a full-length EGFR DMS [19] of ∼22,500 variants across all 1,210 residues scored in the EGFR-dependent human lung cancer line PC9 against erlotinib, afatinib, dacomitinib, and osimertinib (with clinical validation in two glioblastoma patients bearing EGFR G598V); and an EGFR L858R kinase-domain DMS [17], a ∼17,000-variant library built by mutagenic PCR with degenerate codon primer pools (explicitly reporting negligible transition/transversion bias) and scored in Ba/F3 against osimertinib, BLU-945, and seven additional EGFR TKIs. Together with the KRAS G12C and ALK atlases, these chemistry-free DMS atlases demonstrate that chemistry-decoupled DMS of a kinase domain against a clinical inhibitor panel is a reproducible experimental template across kinases and library-construction routes (Twist synthesis, prime editing, MITE-based oligonucleotide tiling, degenerate-codon mutagenic PCR), now extending beyond the KRAS G12C and ALK cases we analyze here.

We decompose a chemistry-biased screen's expected output as a product

$$P(v) \;\propto\; S_{\text{mutagen}}(v) \cdot E(v), \tag{1}$$

with $S_{\text{mutagen}}(v)$ the probability that the given mutagen produces the variant v in one single-nucleotide step from the wild-type codon, and $E(v) = \exp(\sigma(v))$ the drug-induced enrichment factor, with $\sigma(v)$ the drug-selection score read from the atlas on a log scale. On the log scale,

$$\log \hat{P}(v) \;=\; \log S_{\text{mutagen}}(v) \;+\; \beta\sigma(v), \tag{2}$$

with β absorbing any scaling between the atlas's reported units and $\ln E$. For the ALK atlas, $\sigma(v)$ is the Oh atlas Resistance score reported in $\log_2$ units; we work in those units directly (rankings are invariant to the log-base choice, which only rescales β). For the KRAS G12C atlas we treat the Z-score as a dimensionless selection score. All quantitative rank shifts are bracketed by a β-sensitivity interval over $\beta \in [0.25, 4]$ in Results. Some variants have $S_{\text{mutagen}}(v) = 0$, and the two reasons differ in kind. When the required substitution class lies outside the mutagen's chemistry, as with the four classes a cytosine and adenine base-editor pair cannot write, the rate is exactly zero and no library depth recovers the variant. When the wild-type codon is more than one nucleotide away, the one-step rate is zero but the multi-step rate is not, so the variant

is suppressed rather than forbidden; we quantify that suppression in §2.9 instead of asserting impossibility. Both cases are identified without any dependence on β.

Here we apply this decomposition across two case studies. We lead with KRAS G12C, where the inhibitor pipeline is in active development and the predictions are forward-looking: we show that two of twelve variants in a composite resistance panel (patient cohort [9], case report [8], and the accompanying *in vitro* deep mutational scan [9]) lie outside one-step reach from the wild-type KRAS codon, that several more are chemistry-under-counted by current ENU and base-editing screens, and that the previously unidentified Y96D case-report variant [8] is precisely of the kind our framework predicts should be absent or deeply buried in standard ENU/Mutazyme screens. We follow with ALK, where five of the six published Mutazyme-with-alectinib resistance mutations [2] are class-forced and all five land in the Mutazyme-simulated top-120 of the chemistry-free atlas (three in the top-20), as the primary retrospective validation anchor. The framework itself is independent of the three atlases we use for calibration; we expect it to apply to any current or future chemistry-biased screen whose mutagen's substitution spectrum is measured.

2 Methods

2.1 Datasets

KRAS G12C saturation mutagenesis atlas. Average Z-scores for sotorasib and adagrasib across the KRAS G12C protein (positions 2–189) were extracted from Table S1 of the published Twist Biosciences synthetic saturation mutagenesis atlas [10]. The 10-page tables were parsed from the supplementary PDF with `pdfplumber` 0.11.10. The parser enforced that the amino-acid wild-type at each position matched both the KRAS-4A reference and the blank-cell pattern in both the sotorasib and adagrasib blocks. 3,567 variants were recovered per drug, with 188 of 188 positions covered. Position 12 is encoded by the G12C-mutant codon TGT in the library.

ALK saturation mutagenesis atlas. Resistance scores for alectinib, lorlatinib, and zotizalkib, together with fitness scores in the untreated condition, were extracted from Table S2 of the published prime-editing atlas [11], which reports scores for 3,208 single-nucleotide variants across ALK residues 1058–1388 of NM_004304.5, spanning the ALK kinase domain (UniProt 1116–1392) plus flanking exon-boundary and intronic positions. Of these, 2,953 are coding (2,173 missense, 665 synonymous, 115 nonsense) and 255 are intronic (149) or splice acceptor/donor (106); the chemistry-free rankings reported here are computed over the 2,953 coding SNVs after aggregation to the amino-acid level. Drug-selection scores are taken directly from the atlas's Resistance score, $\sigma_{\text{drug}}(v) = \text{Resistance}(v)$, with amino-acid-level aggregation by mean over all synonymous SNV routes to the same amino-acid change. Per the atlas Methods [11], the Resistance score is itself a log fold change comparing drug-treated to untreated cells at the same timepoint ($D10$), so intrinsic variant fitness is already divided out by the common untreated-$D10$ denominator; the atlas's separately reported Fitness score is an orthogonal time-course

quantity ($\log(\text{untreated}_{D10}/D0)$) used by the atlas authors to classify variants on a 2D (Resistance, Fitness) plane ("canonical" / "drug-addiction-like" / "driver-like"). The two scores are empirically near-orthogonal across missense variants ($|r| < 0.05$ for all three drugs); subtracting Fitness from Resistance would therefore double-correct and systematically bias rankings against driver-like variants and in favor of drug-addiction-like variants. We use Resistance alone.

KRAS G12C and ALK resistance panels. The KRAS G12C resistance panel comprises 12 single-amino-acid variants drawn from the clinical and preclinical literature on sotorasib and adagrasib resistance: six patient-observed variants (Y96C, H95D, H95Q, H95R, R68S from a sotorasib-resistance clinical cohort [9]; Y96D from an adagrasib-resistance case report [8]) and six variants reported in the accompanying *in vitro* deep mutational scan [9] at codons 8, 12, 68, 95, and 96 (C12F, V8L, V8M, R68L, R68M, Y96H). Per-variant provenance is given in Supplementary Table S1. We refer to this composite set as the "KRAS G12C resistance panel" throughout and flag patient-observed versus preclinical-scan origin where the distinction affects interpretation. The ALK resistance panel comprises the single-amino-acid alectinib-, lorlatinib-, and zotizalkib-/neladalkib-resistance variants reported in [2, 7], used as the retrospective validation anchor; per-variant provenance is in Supplementary Table S1.

Retrospective validation set (three ALK screens). The retrospective validation in §3.5 and Figure 4 assembles the published variant calls from three preclinical ALK TKI resistance screens: a Mutazyme II alectinib Ba/F3 screen [2] (six single-mutation variants, excluding silent passengers and the I1171S sub-variant of the I1171N/T/S compound call); an ePCR neladalkib compound screen on an I1171N fixed-V1 background [7] (five V2 compound partners reported at 500 nM neladalkib: L1256F, E1129V, E1129K, C1156Y, T1151R); and an industry ENU screen across alectinib, lorlatinib, and neladalkib [14] (thirteen aggregate single-amino-acid variants: I1171N/S/T, F1174C/V, V1180L, T1151K/R, E1129K, R1192P, L1196M, G1269A, G1202R). The industry-screen variant list is aggregated across the three drug arms as the primary report does not disaggregate by arm. For each variant we compute the reachability class, the mutagen's class rate from the mutagen's spectrum (Mutazyme II from the Supplementary NGS of [2]; ENU from [16]), and the prime-editing atlas chemistry-free and simulated ranks. Because the prime-editing atlas does not include neladalkib, the two neladalkib arms use zotizalkib as a 4th-generation ALK TKI proxy for the σ term; this is flagged in the figure and in the derived data file `retrospective_validation.csv`. R1192P falls outside the atlas's residue window (1058–1388) and is reported as unanchored.

2.2 Codon tables

The ALK NM_004304.5 codon table was assembled from ClinVar variant records anchored to NM_004304.5 at each residue reported in the ALK resistance panel [2, 7, 11], and cross-checked against the EML4-ALK v1 wild-type CDS in the ALK atlas supplement. The KRAS-4A NM_033360.4 codon table was reconstructed from the canonical protein sequence (verified

against the blank-cell pattern in the KRAS atlas table) and ClinVar-anchored at the resistance-panel residues (V8, G12/C12, S17, Q61, R68, H95, Y96). The 4A-specific hypervariable region (positions 151–189) was placed by consensus and is not individually ClinVar-verified, but this region is not involved in the resistance panel. Both tables are distributed as YAML files with per-residue `verified_by` annotations.

2.3 Mutagen substitution spectra

The Mutazyme II spectrum is taken from the supplementary NGS of the untreated library in [2]. The ENU nominal spectrum is the compilation [16] of ENU DNA-adduct conversion frequencies across published mouse germline mutagenesis studies [16], re-expressed in the six-class pyrimidine convention. That compilation is phenotype-ascertained: its 587 substitutions are the ones that produced a scorable phenotype in viable mice, so they have passed both an embryonic-viability filter and a phenotype-detection filter, and they come from mouse germline rather than from a Ba/F3 culture. Both differences push in unknown directions. No pre-selection six-class spectrum measured on an *in vitro* ENU-treated mammalian library exists in the published literature, and we were unable to locate one. We therefore use this compilation as a nominal reference rather than as a calibrated pre-selection spectrum, we report its sensitivity wherever it enters a quantitative claim, and §3.5 tests it against the one *in vitro* ENU output we can observe and finds it inconsistent. The base-editing nominal spectrum assigns equal mass to C>T and T>C and zero to the other four classes, modeling a 1:1 CBE:ABE mix at the editable positions. The uniform chemistry-free baseline assigns 1/6 to each of the six classes. All spectra are stored as six-element vectors summing to 1 and pass a validation that enforces non-negativity, class completeness, and tolerance of 5% on the sum.

2.4 Class-forced classification

For each (wild-type amino acid, mutant amino acid, wild-type codon) triple, we enumerated the single-nucleotide-variant routes from the wild-type codon to any codon encoding the mutant amino acid, and assigned each route to one of the six pyrimidine substitution classes using the standard reverse-complement convention. A (wild-type, mutant) pair is *reachable in one nucleotide* if at least one SNV route exists, and *class-forced* if every such route belongs to the same pyrimidine class. The enumeration is performed per-codon when the wild-type codon is known; otherwise it is performed over every synonymous codon encoding the wild-type amino acid and the result is reported as class-forced only when every codon's routes agree on a single class.

2.5 Decomposition and ranking

For each class-forced variant v with forced class $c(v)$ and drug-selection score $\sigma_{\text{drug}}(v)$ from the chemistry-free atlas, the simulated preclinical log-score under mutagen m follows Eq. 2 at $\beta = 1$: $\log \hat{P}_{\text{simulated}}(v, m, \text{drug}) = \log S_m(c(v)) + \sigma_{\text{drug}}(v)$, and the chemistry-free log-score is

$\log \hat{P}_{\text{chemfree}}(v, \text{drug}) = \sigma_{\text{drug}}(v)$. Rank-shift point estimates throughout Results are reported at $\beta = 1$ and bracketed by intervals over $\beta \in [0.25, 4]$ to make the selection-scale dependence explicit. Variants are ranked in decreasing order under each score; the "rank shift" is $\text{rank}_{\text{simulated}} - \text{rank}_{\text{chemfree}}$. A positive rank shift means the mutagen under-counts the variant relative to the chemistry-free baseline; a negative shift means it over-counts.

Figure 3B reports the top-K recovery curve of each (gene, drug, mutagen) combination against the chemistry-free ground truth. For each $K \in [1, N]$ with N the class-forced saturation-atlas size, we compute

$$\text{recovery}(K; m, \text{drug}) \;=\; \frac{\left|\, \text{top}_K(\log \hat{P}_{\text{simulated}}) \,\cap\, \text{top}_K(\log \hat{P}_{\text{chemfree}}) \,\right|}{K}. \tag{3}$$

Variants with $S_m(c(v)) = 0$ (chemistry-unreachable under m) are retained at a tied-bottom simulated rank by substituting $S_m \to 10^{-6}$ before log-transforming, so that recovery reflects both within-coverage ranking distortion and coverage loss across unreachable variants. The chemistry-free reference curve (Twist [10] for KRAS G12C, prime editing [11] for ALK) is identically $y = 1$ by construction because the uniform class supply $S_m \equiv 1/6$ makes $\log \hat{P}_{\text{simulated}}$ differ from $\log \hat{P}_{\text{chemfree}}$ by an additive constant and preserves the ranking. Curves in Figure 3B are smoothed with a rolling 5-point median to tame small-K noise without erasing the underlying step structure.

2.6 Library-depth detection probability

The decomposition in Eq. 1 is a one-shot supply model. Real screens sample a library of N cells with per-base mutation rate μ, and under drug selection a single resistant clone can expand to detection. The probability that a given variant v has at least one carrier in a library of size N is $P_{\text{detect}}(v) \approx 1 - \exp\big(-N\,\mu\,S_m(c(v))\,r(v)\big)$, with $r(v)$ the number of single-nucleotide routes from the wild-type codon to v. For a Ba/F3 library of $N = 10^7$ cells and $\mu \sim 10^{-4}$ per base per round, P_{detect} saturates near 1 for any supplied class above $\sim 10^{-4}$. Every class in every chemical spectrum used here clears that bound, including ENU's C>G at 2×10^{-3}. Rank-shift predictions therefore apply in the depth-limited regime, and at practical library depth the binary distinction $S = 0$ versus $S > 0$ is the operational one. That distinction is where the reachability results of this paper bind regardless of β: exactly for the four classes base editing cannot write, and to within the two-order suppression computed in §2.9 for the two-step variants R68L and V8M.

2.7 Benchmark against null rankings

For each retrospective anchor (Doi et al. 2022 ALK Mutazyme × alectinib [2]; Tangpeerachaikul et al. 2025 industry ALK ENU aggregate [14]; Awad et al. 2021 KRAS G12C clinical cohort under Mutazyme-simulated sotorasib and adagrasib [9]), we compute three rankings over the class-forced universe of the relevant atlas: σ alone (chemistry-free); the framework $\log S + \sigma$;

and supply alone $\log S$. For each ranking we report top-10, top-50, and top-100 recall, and AUROC (Table 2). Permutation p-values for the framework-minus-σ-alone top-100 recall gain are computed by shuffling the *forced_ class* labels across variants and recomputing the framework ranking 5,000 times with a fixed seed (`numpy` rng seed 20261013); the empirical two-sided p-value is $(\sum \mathbb{1}\{|\text{null gain}| \geq |\text{observed gain}|\} + 1)/(n_{\text{iter}} + 1)$.

These anchors are small, so we also report the resolution of the permutation test. Recall at n observed hits moves only in units of $1/n$, which sets a floor on any observable effect, and we take the 97.5th percentile of the permutation null's absolute gain as the smallest gain that would reach two-sided significance. Both quantities appear in Table 2. A non-significant p above that threshold bounds the chemistry term's contribution; it is not evidence that the contribution is zero. The Oh et al. prime-editing atlas does not include neladalkib, so for the industry anchor (R10) we proxy σ by each of alectinib, lorlatinib, and zotizalkib in turn. Table 2 reports all three proxy results; no proxy gives a framework gain over σ-alone that reaches conventional significance.

The agreement between any ranking and the atlas's true σ is bounded by the atlas's own replicate-level reproducibility. The Oh 2026 [11] atlas reports replicate Pearson $r = 0.67$–0.72 at SNV level across the three drugs; by Spearman–Brown, the two-replicate reliability is ≈ 0.82, so the maximum attainable correlation between any model's ranking and the underlying resistance score is $\sqrt{0.82} \approx 0.91$. Top-K overlap ceilings sit strictly below this correlation ceiling. The 63%–72% top-100 recovery for the SNV mutagens reported here therefore cannot be read as a quantitative deficit of the biased chemistries without reference to this ceiling; a formal rep-vs-rep top-K ceiling requires access to the per-replicate atlas scores, which are not released in Table S2 and are deferred to a follow-up when the authors' repository is available.

2.8 Why hit-class composition needs a post-selection null

A screen's reported variants have passed drug selection, so their substitution-class composition is a post-selection quantity. Comparing it with the mutagen's pre-selection spectrum attributes to supply any class enrichment that selection produced. We therefore evaluate an observed count of k hits of class c in n hits against five nulls for $\pi(c)$, the probability that a hit belongs to class c:

N0, raw spectrum. $\pi(c) = S_{\text{mutagen}}(c)$. This is the comparison that ignores selection.

N1, supply × composition. $\pi(c) \propto S_{\text{mutagen}}(c)\, n_c$, with n_c the number of class-forced atlas variants in class c. Corrects for unequal class sizes only.

N2, selection only. $\pi(c)$ is the share of class c among the top-N σ-ranked class-forced variants ($N = 50, 100, 200$), with no supply term. This is the composition a chemistry-free screen would report.

N3, model-implied. $\pi(c) \propto \sum_{v \in c} S_{\text{mutagen}}(c) \exp(\beta\sigma(v))$, that is, Eq. 2 used as its own null. Under the hypothesis that the reference spectrum describes the library, N3 is the hit-class

distribution the framework predicts.

N4, atlas composition. $\pi(c) = n_c / \sum_{c'} n_{c'}$, the share of class c among all class-forced atlas variants. Drops supply and selection both, leaving only how many variants each class contains.

N3 is the test of the spectrum; N2 is the test of whether chemistry enters at all; N4 is the floor against which both are read. We report $P(X \geq k)$ under each, with σ pooled across drug arms as $\max_d \sigma_d(v)$, since a variant enters an aggregate hit list if it is selected in any arm, and we repeat the calculation per arm and across $\beta \in [0.25, 4]$. Inverting an observation to an implied supply rate is only licensed under N3: we solve $\pi_{\mathrm{N3}}(c; s) = \hat{p}$ for the spectrum in which class c carries mass s and the other five keep their reference ratios rescaled to $1 - s$.

2.9 Multi-nucleotide reachability

A mutagen applies its per-base rate to every base, so a variant requiring two substitutions within one codon is suppressed rather than forbidden. For a route whose two substitutions fall in classes c_1 and c_2, the expected number of carriers acquiring both in one round is $N\mu^2 S(c_1)S(c_2)$ summed over routes and both orderings; over R rounds of mutagenesis and expansion, sequential acquisition through a single-mutant intermediate contributes a further factor of order $R(R+1)/2$. Routes are enumerated from the genetic code rather than assumed. We report expected carriers and $1 - \exp(-\lambda)$ across $\mu \in [10^{-5}, 10^{-3}]$ and $R \in \{1, 5, 20\}$, and express the result as fold suppression relative to a one-step variant of comparable class supply in the same library.

2.10 External test of σ transport

The atlases measure drug response in H3122 cells at mild selection, while the screens we audit run in Ba/F3 at 5–15× IC_{50}. To test whether σ survives that change of system, we compare it against measured Ba/F3 cellular IC_{50} values compiled from the published ALK inhibitor literature across seven studies [20–26]. We keep only EML4-ALK v1 Ba/F3 cellular IC_{50} measurements of single amino-acid substitutions, excluding compound mutants and indels. Fold resistance is computed within each (study, drug) stratum against that study's own wild-type measurement, so differences in absolute IC_{50} between laboratories cancel; variants measured more than once are summarized by the median $\log_{10}$ fold change, and the inter-study range is retained as a noise estimate. Agreement is Spearman rank correlation, since every use of σ in this paper is a ranking, with a 10,000-sample bootstrap interval over variants. The same comparison against measured neladalkib potency ranks the three candidate σ proxies for the drug the atlas does not contain.

The internal test uses the atlas's own multi-dose sub-library tables (Additional file 2, Tables S3, S5 and S9 of [11]), which assay exon-23 and exon-22 variants at two or three concentrations per drug. We keep missense rows, average duplicate pegRNAs targeting the same amino-acid change, and compare concentrations by Spearman correlation and by top-25 set overlap. Because a bare dose-to-dose correlation admits several readings, we run four controls: which concentration

the main single-dose atlas resembles, whether different drugs assayed at matched dose rank agree better than at unmatched rank, whether score dispersion grows with concentration, and whether the sub-library window is range-restricted relative to the full atlas. The dose columns are labelled in µM in the workbook while the published Methods give nanomolar concentrations for alectinib and lorlatinib; we therefore report dose ratios, which are invariant to that discrepancy, rather than absolute concentrations.

2.11 What each atlas can and cannot contain

"Chemistry-free" means uniform across the six substitution classes. It does not mean unrestricted in reach, and the three atlases differ on that second axis. The Oh ALK atlas is built by prime editing from a pegRNA library of single-nucleotide edits: all 2,173 of its kinase-domain missense variants are reachable in one nucleotide step, so it holds no two-step variants and cannot measure that blind spot. The Feng KRAS G12C Twist library and the Estevam MET scan are synthesized or assembled rather than edited, and both contain variants that no single substitution can produce, which is why R68L and V8M appear in the KRAS panel at all. Quantities that depend on two-step variants are therefore reported from the KRAS and MET pipelines only.

2.12 MET atlas, codon table and hit lists

$\sigma(v)$ for MET is the Estevam ROSACE effect under a given inhibitor minus the effect in that study's DMSO arm [18]. The subtraction is what makes this quantity the same as the ALK one rather than a departure from it: Oh's Resistance score is already log(drug/untreated), so growth is divided out before we see it, which is why its Fitness axis is orthogonal and is not subtracted again; Estevam's score is computed per arm against that arm's own starting pool, so the drug arms still carry growth, and the DMSO arm correlates with them at $r \approx 0.8$ rather than ≈ 0. The difference reproduces Oh's quantity, and matches the resistance definition used in that study's own analysis code. The published tables number the kinase domain locally from 1 to 287; we re-derive the offset to canonical numbering from the data by scoring every candidate against known MET residue identities, which selects +1058 uniquely (10 of 10 anchors, next best 2 of 10), giving residues 1059–1345.

The MET codon table is read from RefSeq NM_000245.2, the record Tiedt et al. state their numbering refers to, and is accepted only after three checks pass: the translated CDS equals the record's own protein, every one of the 287 atlas positions encodes the wild-type residue the atlas reports, and all 25 reported nucleotide alterations fall in the right codon, start from the base the table holds and produce the amino acid reported. Hit lists and clone counts are transcribed from Supplementary Tables S1 and S2 of [12] and cross-checked against that paper's stated totals of 85 and 60 clones, 10 and 15 distinct nucleotide alterations, and 4 and 10 affected residues.

2.13 Data provenance and parsing validation

Every derived table in `data/derived/` is produced by a named script in `scripts/` or a numbered notebook in `notebook/`; the `paper/NUMBERS.md` master book links every quantitative claim in Results to the specific function and input file that produced it. The parsed Feng Table S1 and Oh Table S2 long-format CSVs are redistributed with SHA-256 checksums collected in `CHECKSUMS.sha256`, regenerated from `sha256sum.py` on every build. The pdfplumber extraction of the Feng supplementary PDF is self-validated by two integrity constraints enforced at parse time: (i) the amino-acid wild type at each position must match the KRAS-4A reference, and (ii) the blank-cell pattern (the self-identity row) must appear at every residue in both the sotorasib and adagrasib blocks. Both constraints are tight: a single mis-aligned row violates one or both and aborts the parse. 3,567 variants were recovered per drug with no violations across 188 of 188 positions. The ALK atlas scores in [11] are published as log fold changes; following standard CRISPR-screen convention and the atlas's own supplementary code, we treat these as $\log_2$ values. Rankings under the chemistry-free atlas are invariant to the choice of log base (which absorbs into β in Eq. 2), so this choice does not affect any rank-shift or top-K recall number in Results.

2.14 Software and reproducibility

All analysis is in Python 3.13.16 with pandas 3.0.5, numpy 2.5.3, scipy 1.18.1, matplotlib 3.11.2, and pyyaml 6.0.3; this stanza is emitted from the running interpreter by the script `env_versions.py` rather than recorded by hand. Figures were rendered at 300 dpi with matplotlib. Codon tables, spectra, and derived variant tables are released as plain text (YAML and CSV); figure-generation code is in the `paper/` directory of the repository. The random seed used for all stochastic steps is `20261013`. Haldane–Anscombe pseudocounts (0.5) are applied to any count-based estimator to avoid log-zero.

3 Results

3.1 Current preclinical mutagens span four distinct coverage tiers

We first cataloged the mutagens used in the ALK and KRAS G12C resistance screens published between 2022 and 2026 (Fig. 2B). Three distinct chemistries dominate current practice: ENU is used in an industry screen for the fourth-generation ALK TKI neladalkib [14] and in a KRAS G12C screen for sotorasib and adagrasib [3]; error-prone PCR (Mutazyme-class) is used in the next-generation ALK TKI panel screen [7]; and CRISPR base editing (CBE+ABE) is used in a KRAS G12C screen for adagrasib and the RAS(ON) tri-complex inhibitors [15]. Only two studies in the window used chemistry-free libraries: a Twist synthetic library for KRAS G12C [10] and a prime-editing atlas for ALK [11].

No chemistry in current practice supplies the six substitution classes uniformly (Fig. 2A; the five studies are listed in Table 1). ENU is dominated by T>C (36.6%) and T>A (31.4%), with

C>G at only 0.2%; Mutazyme II is dominated by C>T (47.0%) and C>A (18.0%), with T>G at 3.6%; CBE+ABE is restricted by design to C>T and T>C and supplies 0% at the four other classes. No class exceeds 50% of any mutagen's output. The uniform baseline ($1/6 \approx 16.7\%$ per class) is reached only by Twist or prime editing.

Class coverage partitions the four chemistries into four tiers (Fig. 2B). (i) Twist and prime editing supply all six classes uniformly. (ii) Mutazyme II supplies all six classes at $\geq 1\%$ but with the rarest class at 3.6% and the most abundant at 47.0%, a 13-fold ratio. (iii) ENU supplies five of six classes at $\geq 1\%$, with C>G reduced to 0.2%. (iv) Base editing (CBE+ABE) supplies only two of six classes. Only one of these ceilings is structural. For CBE+ABE the four unwritten classes carry an exactly zero rate, so a variant whose single-nucleotide route falls in one of them is not producible at any library depth or selection pressure. For ENU and Mutazyme II the ceiling is quantitative, and library depth decides whether it binds at all. ENU's rarest class, C>G at 0.2%, still yields 2.0 expected carriers per route in a 10^7 library at $\mu = 10^{-4}$, a detection probability of 0.86 (§2.6). A rare class is therefore supplied, not excluded, and a 13-fold internal class ratio shifts expected rank order without removing anything from the output. The KRAS G12C H95D case in §3.3 illustrates the quantitative ceiling, and §3.5 shows how little of the observed hit composition it accounts for.

Table 1. Preclinical TKI-resistance screens published 2022–2026 and the mutagen chemistry each uses. The five studies cover both kinases of interest here; the three most recent (industry neladalkib ENU, next-gen ALK ePCR, KRAS G12C CBE+ABE) all employ chemistry-biased mutagens.

Study	Gene / drug(s)	Mutagen	Classes supplied
Nuvalent 2025 [14]	ALK (neladalkib)	ENU	5 of 6, biased
Takei 2025 [7]	ALK (alec/lor/zoti/nela)	error-prone PCR	6 of 6, biased
Diaz 2025 [15]	KRAS (ada, RMC-4998, RMC-7977)	CBE + ABE	2 of 6
Oh 2026 [11]	ALK (alec/lor/zoti)	prime editing	6 of 6, uniform
Feng 2022 [10]	KRAS G12C (sot/ada)	Twist synthetic	6 of 6, uniform

3.2 A structural blind spot in the KRAS G12C resistance panel

We assembled a composite KRAS G12C resistance panel of twelve single-amino-acid variants from the clinical and preclinical sotorasib- and adagrasib-resistance literature: six variants reported in patient tumors (Y96C, H95D, H95Q, H95R, R68S from a clinical cohort of 11 patients with sotorasib-emergent resistance [9]; Y96D from a clinical case report of adagrasib resistance [8]) and six variants reported in the accompanying *in vitro* deep mutational scan [9] at codons 8, 12, 68, 95, and 96 (C12F, V8L, V8M, R68L, R68M, Y96H). Per-variant provenance is listed in Supplementary Table S1. Using a ClinVar-anchored KRAS-4A codon table (Methods), we classified each variant by its single-nucleotide reachability from the wild-type codon in the G12C-mutant background: 2 of 12 variants (R68L, V8M) require 2 or more nucleotide changes and so lie outside one-step reach for every mutagen; 3 of 12 (V8L, R68S, H95Q) are reachable in a single nucleotide via two distinct pyrimidine classes and so cannot be assigned a single "forced class" by amino-acid grammar; and 7 of 12 (C12F, Y96C, Y96D, Y96H, H95D, H95R, R68M) are

reachable in a single nucleotide via exactly one class and are the variants to which the chemistry decomposition in Equation 1 applies directly (Fig. 1B).

The two-step category includes R68L, the top-ranked variant in the chemistry-free KRAS G12C atlas under both sotorasib (Z = +3.47) and adagrasib (Z = +4.34). The wild-type R68 codon is AGG, and the shortest nucleotide path from AGG to any leucine codon requires two base substitutions, by way of CTG or TTG. V8M is the same case with one route rather than two (GTA → ATG). Both variants are reported in the accompanying *in vitro* deep mutational scan [9], whose library construction admits multi-nucleotide events, and neither appears in that study's 11-variant clinical cohort (G12D, G12R, G12V, G12W, G13D, Q61H, R68S, H95D, H95Q, H95R, Y96C). Their warrant is therefore preclinical rather than clinical, and the claim we make about them is a codon-level one.

Two substitutions in one codon are suppressed rather than forbidden, and the paper's own compound-mutation anchor [7] is an error-prone PCR screen whose results depend on multi-nucleotide events happening. We therefore quantify the suppression (§2.9). In a 10^7 library at $\mu = 10^{-4}$ per base per round, a single round of mutagenesis yields 0.005–0.017 expected carriers of R68L or V8M, a presence probability below 2%. Twenty rounds raise the expected count to 1.1–3.6 carriers, at which point both become likely present but remain 51- to 169-fold below a one-step variant of comparable class supply in the same library. The operational claim is that two-step variants are suppressed by roughly two orders of magnitude at practical depth, and that single-round protocols, which include the standard ENU Ba/F3 treatment, will usually not contain them at all. Consistent with this, an ENU-based Ba/F3 KRAS G12C screen reporting only Y96D and Y96S as hits [4] is silent on R68L despite its position at the top of the chemistry-free ranking.

The twelve-variant panel is the set of variants screens have reported, so the blind spot it displays could in principle be a property of that set rather than of the chemistry. It is not. Extending the same calculation to every residue of all three kinase domains, and asking at each one whether the chemistry can make the substitution the chemistry-free atlas ranks highest there, base editing is blocked at 172 of 273 ALK residues, 121 of 188 KRAS G12C residues and 183 of 287 MET residues, a share of 63%–66% that holds across all three genes and every drug arm (Fig. 3C). ENU and Mutazyme II are blocked at no residue in any of them, which is the threshold-not-weight claim in its plainest form: the two chemistries that supply all six classes forbid nothing, and the one that supplies two forbids two residues in three. Because the three atlases were built by prime editing, synthesis and deep mutational scanning respectively, the agreement is not an artifact of any one library design.

3.3 Chemistry restructures the KRAS G12C ranking by orders of magnitude

For the 7 of 12 class-forced variants in the resistance panel, we computed the simulated preclinical rank under ENU and Mutazyme II supply and compared it to the chemistry-free rank derived from the KRAS G12C saturation atlas directly (Fig. 1C, Fig. 3). We examine the two most

under-counted variants in the panel in detail here: H95D under adagrasib and Y96D under sotorasib.

H95D is reported in the clinical cohort [9] as an adagrasib-emergent resistance variant, forced class C>G, chemistry-free rank 9 of 981 class-forced variants. The simulated rank-shift depends on the selection-scale parameter β in $\log \hat{P} = \log S + \beta\sigma$ (Methods, Eq. 2). Under Mutazyme II, whose NGS-measured C>G rate is 5%, the simulated rank is 56 and the shift is +47 at $\beta = 1$, with an interval of +1 to +606 across $\beta \in [0.25, 4]$. Under the germline-ENU reference [16], which places C>G at 0.2%, the shift is +15 to +848 over the same range. We do not report a third row using an *in vitro*-calibrated ENU C>G rate. The earlier draft of this analysis derived one by inverting the industry anchor, and §3.5 shows that inversion is not identified: the same observation is equally consistent with a higher supply rate and with supply not controlling the hit list at all. The width of the Mutazyme interval is the honest summary of what a class-rate difference does to rank here, and it spans the full atlas.

For Y96D, the Mutazyme II simulated ranking under sotorasib places the variant at position 139 (chemistry-free position 7), a +132-rank shift at $\beta = 1$ (interval +5 to +643 across $\beta \in [0.25, 4]$). The forced class for Y96D is T>G, which Mutazyme II supplies at 3.6% [2]. The primary clinical report of Y96D [8] describes it as "previously unidentified"; prior preclinical screens had used ENU or Mutazyme II and had reported Y96C (T>C, well-supplied) but not Y96D (T>G, under-supplied). That pattern is consistent with a chemistry contribution to the clinical identification delay, though it does not establish one. A subsequent Twist-complemented preclinical screen [4] reports Y96D and Y96S.

The full reachability matrix (Fig. 3A) partitions the KRAS G12C and ALK resistance panels into three screen-design categories at a glance. For each (variant, mutagen) cell, the matrix shows the maximum class-supply probability achievable by that mutagen for that variant. Cells colored in blue are chemistry-accessible (supply $\geq$ 10%); cells in amber are chemistry-under-counted (supply $<$ 10%); cells in red are two-step from the wild-type codon. The KRAS G12C half of the matrix shows that CBE+ABE cannot produce 8 of 12 panel variants (0% supply): C12F, Y96D, H95D, H95Q, R68S, and R68M, together with the two two-step variants (R68L, V8M). Four of the eight are patient-observed: Y96D [8], and H95D, H95Q, and R68S [9]. The other four, including R68L and V8M, are reported only in the accompanying *in vitro* scan.

3.4 The ALK atlas reproduces the published alectinib resistance panel

We applied the same decomposition to the ALK kinase domain, using the Oh et al. [2026] prime-editing atlas as $\sigma(v)$ and the Mutazyme II spectrum as $S_{\text{mutagen}}(v)$ (Fig. 1D). Across the 1,427 class-forced ALK kinase-domain variants, the fraction of the chemistry-free top-100 priorities that the Mutazyme-simulated screen also places in its top 100 is 65% (Wilson 95% CI 55%–74%) for alectinib, 72% [63%–80%] for lorlatinib, and 65% [55%–74%] for zotizalkib; ENU performs comparably (63% [53%–72%], 65% [55%–74%], 63% [53%–72%] respectively). For the 981 class-forced KRAS G12C variants the same SNV-mutagen top-100 recoveries sit at 40%–54% across

the two drugs and the two chemistries (individual Wilson 95% half-widths ≈ 10 pp). ENU is dominated by T>C/T>A and Mutazyme II by C>T, with internal class-rate ratios of 183-fold and 13-fold respectively. Despite those markedly different six-class spectra, the two mutagens give near-identical top-100 recovery across the three ALK drugs (63%–72%, with per-drug same-chemistry CIs overlapping between ENU and Mutazyme II for every drug). For SNV chemistries, chemistry's effect on ranking is second-order; the dominant failure mode is coverage loss, not rank reordering. Figure 3B shows the full top-K recovery curves (with Wilson 95% CI bands) for all five (gene, drug) combinations.

Base editing fails in a qualitatively different way. The CBE+ABE curves in Fig. 3B (purple) show that, across the three ALK drugs, base editing recovers only 0%, 10%, and 10% of the chemistry-free top-10 variants for alectinib, lorlatinib, and zotizalkib respectively, which leaves 90%–100% of each drug's top-10 outside its reach, and its peak recovery across $K \leq 100$ reaches 38% for ALK and 50% for KRAS G12C. Its inability to supply four of six substitution classes sets that ceiling.

An earlier Mutazyme II Ba/F3 ALK alectinib screen [2] reported several resistance mutations without quantitative frequencies or an explicit ranking. In our simulation, five of the six reported mutations are class-forced; three of the five (L1196M, L1198F, V1180L) sit in the Mutazyme-simulated top-20 of 1,427 class-forced ALK variants, and all five sit within the top-120 (per-mutation provenance in Supplementary Table S1). In particular L1196M (class C>A, Oh atlas Resistance score for alectinib +7.7) appears at rank 1 of 1,427 under both chemistry-free σ and the Mutazyme-simulated ranking, L1198F (class C>T, Mutazyme's dominant class at 47%, Resistance score +4.6) moves from chemistry-free rank 18 to Mutazyme rank 3 as the C>T weighting pulls it up, and V1180L (class C>G, under-supplied at 5% in Mutazyme, Resistance score +5.3) sits at chemistry-free rank 8 and simulated rank 15. The sixth reported variant, G1202R, is multi-class SNV-reachable and so sits outside the class-forced ranking; it is recovered on its own SNV paths. Two chemistry-sensitive variants in our ALK-with-alectinib panel that are not in that screen's reported set are C1156Y (chemistry-free rank 132, simulated 55, moved up because its forced class C>T is Mutazyme's dominant class) and L1152R (chemistry-free rank 225, simulated 822, moved down because its forced class T>G is under-supplied). Both shifts have the sign and approximate magnitude predicted by the Mutazyme spectrum.

Taken together, the KRAS G12C and ALK results support the same conclusion from independent atlases and distinct clinical contexts: chemistry restructures preclinical rankings in a predictable way, and some clinically important variants are completely outside what any single-nucleotide chemistry can produce.

3.5 Validation across three preclinical ALK screens

We stress-tested the decomposition against two additional preclinical ALK TKI resistance screens published since 2022: an error-prone PCR screen on an I1171N fixed-V1 background with neladalkib selection [7], and an industry ENU screen across alectinib, lorlatinib, and neladalkib

[14] (Fig. 4; derived data in `retrospective_validation.csv`). Together with the Mutazyme II-alectinib screen [2] already shown (§3.4), this gives three independent ALK screens spanning three mutagens, four ALK TKIs, and both the single-mutation and compound-mutation regimes. Only two of them enter the benchmark below, because the compound screen supplies no single-mutation σ to rank against.

We assembled four retrospective anchors: Doi et al. 2022 [2] (ALK Mutazyme II × alectinib, five class-forced hits); Tangpeerachaikul et al. 2025 [14] (industry ALK ENU screen, eleven class-forced hits aggregated across alectinib, lorlatinib, and neladalkib arms); and the Awad et al. 2021 clinical cohort [9] analyzed under Mutazyme II for sotorasib and adagrasib (four class-forced patient-observed KRAS G12C hits under adagrasib and three under sotorasib). On each anchor we compared three rankings over the chemistry-free atlases: σ alone, the framework $\log S + \sigma$ at $\beta = 1$, and supply alone $\log S$ (Table 2). σ alone recovers 60%–100% of observed hits in the top-100 (AUROC 0.77–0.99); adding the chemistry term as $\log S + \sigma$ changes the top-100 recall by at most ±33 pp (the framework ties σ-alone on all four ALK anchors and falls below it on both Awad anchors) and the gain is not statistically significant by class-label permutation on any anchor at the conventional threshold ($p = 1.00$ on each of the four ALK anchors; $p = 0.30$ and 0.60 for the Awad sotorasib and adagrasib anchors; 5,000 shuffles). Supply alone recovers 0%–20% across all four single-mutation anchors and every proxy choice.

Table 2. Benchmark: σ-alone, framework ($\log S + \sigma$ at $\beta = 1$), and supply-alone rankings on four retrospective anchors. Columns labeled Top-K give the top-K recall, the fraction of observed hits captured in the top K of each ranking. N is the class-forced universe size (981 for KRAS G12C, 1,427 for ALK). p is a class-label permutation p-value for the framework-minus-σ-alone gain at $K = 100$ (5,000 shuffles, two-sided). MDG is the minimum detectable gain, the smallest absolute top-100 recall gain that would reach two-sided significance at that anchor's size; every observed gain lies below it, so these p-values bound the chemistry term rather than nullify it. The chemistry weighting at $\beta = 1$ does not improve recall over σ alone at any resolution these anchors can supply. For the industry anchor, the Oh et al. atlas does not report neladalkib, so we proxy σ with each of alectinib, lorlatinib, and zotizalkib and tabulate all three; the conclusion that the chemistry term does not consistently improve over σ alone survives every proxy choice. Measured Ba/F3 neladalkib potency ranks those proxies (§3.6): lorlatinib and zotizalkib perform equally and alectinib performs worse.

Anchor	Ranking	n	Top-10	Top-50	Top-100	AUROC	p	MDG
Doi (ALK Mut, alec)	σ alone	5	0.40	0.40	0.60	0.94	—	—
	$\log S + \sigma$	5	0.20	0.40	0.60	0.86	1.00	0.20
	$\log S$ alone	5	0.00	0.20	0.20	0.48	—	—
Industry (ALK ENU)*	σ alone (alec)	11	0.27	0.64	0.82	0.94	—	—
	$\log S + \sigma$ (alec)	11	0.27	0.64	0.82	0.93	1.00	0.18
	σ alone (lor proxy)	11	0.36	0.73	0.91	0.94	—	—
	$\log S + \sigma$ (lor proxy)	11	0.27	0.82	0.91	0.94	1.00	0.18
	σ alone (zot proxy)	11	0.09	0.64	0.64	0.77	—	—
	$\log S + \sigma$ (zot proxy)	11	0.00	0.64	0.64	0.76	1.00	0.09
	$\log S$ alone (zot proxy)	11	0.00	0.00	0.09	0.42	—	—
Awad (KRAS Mut, sot)	σ alone	3	0.67	0.67	0.67	0.93	—	—
	$\log S + \sigma$	3	0.33	0.33	0.33	0.88	0.30	0.33
	$\log S$ alone	3	0.00	0.00	0.00	0.50	—	—
Awad (KRAS Mut, ada)	σ alone	4	0.50	1.00	1.00	0.99	—	—
	$\log S + \sigma$	4	0.25	0.50	0.75	0.92	0.60	0.50
	$\log S$ alone	4	0.00	0.00	0.00	0.44	—	—

* The prime-editing atlas does not include neladalkib, so each of alectinib, lorlatinib, and zotizalkib is used as a 4th-generation ALK TKI σ proxy; all three are tabulated.

The chemistry term, as a quantitative modifier of σ, does not improve rank prediction over σ alone at $\beta = 1$ on any anchor. The strength of that statement is set by the anchors, which hold 3 to 11 observed hits. Recall moves only in whole hits, so the smallest observable change is 9 to 33 pp depending on the anchor, and the permutation tests resolve gains of 9 to 50 pp (Table 2). Every observed gain falls inside that window. These anchors therefore exclude a large contribution from the chemistry weighting and say nothing about a small one.

What the chemistry term does contribute is structural. Supplying zero of a required substitution class removes a variant from the output entirely: R68L and V8M in KRAS G12C, and the four of six classes base editing does not supply. Within the classes a given mutagen does supply, drug selection as reported in a chemistry-free atlas is the dominant signal.

Primary anchor (Doi 2022). A Mutazyme II Ba/F3 alectinib screen [2]: five of six observed resistance variants are class-forced and all five fall in the Mutazyme-simulated top-120 of 1,427 (L1196M rank 1, L1198F rank 3, I1171T rank 10, V1180L rank 15, I1171N rank 118); the sixth, G1202R, is multi-class SNV-reachable.

Compound-mutation anchor (Takei 2025). An error-prone PCR neladalkib screen on an I1171N fixed-V1 background [7] recovers five V2 partners (L1256F, E1129V, E1129K, C1156Y, T1151R); three of the five are class-forced into C>T, Mutazyme II's dominant class at 47% of substitutions. The compound-mutation σ required to rank these as single variants is not provided by a single-mutation atlas, so this anchor contributes a class-rate consistency observation rather than a top-K precision entry in Table 2.

Industry ENU anchor (Nuvalent 2025). An ENU screen across alectinib, lorlatinib, and neladalkib [14] reports thirteen single-amino-acid variants aggregated across its three drug arms. Twelve of thirteen are class-forced; R1192P is outside the atlas residue window. Nine of twelve class-forced hits fall in classes ENU supplies at $\geq 1\%$. The remaining three (V1180L, T1151R, G1269A) are class-forced into C>G, which the germline-ENU compilation [16] places at 0.2%.

These thirteen variants are post-selection: they appear in the hit list because they survived drug selection. Testing their class composition against the pre-selection spectrum would charge to supply what Eq. 1 assigns to selection, so we test the count against five nulls instead (Methods, §2.8; Table 3). Under the model-implied null, in which the germline spectrum sets supply and $\exp(\beta\sigma)$ sets selection, selection raises the expected C>G share of hits from 0.2% to 0.56% at $\beta = 1$. Three of thirteen remains improbable under that null ($P = 4.7 \times 10^{-5}$ at $\beta = 1$, which is the least extreme value across $\beta \in [0.25, 4]$; the tail falls to 4.1×10^{-10} at $\beta = 4$), so the germline spectrum does not describe this screen's output.

The same count carries no information about the chemistry weighting. C>G variants are enriched among high-σ ALK variants: 24 of the top-100 σ-ranked class-forced variants are C>G, against 243 of 1,427 (17.0%) in the atlas as a whole. Against a selection-only null, three C>G hits in thirteen is unremarkable ($P = 0.64$), as it is against atlas composition alone ($P = 0.39$). Two

readings therefore survive: the *in vitro* ENU C>G rate exceeds the germline reference, or the class weighting does not control which variants reach the hit list. The depth calculation separates them. At $N = 10^7$ and $\mu = 10^{-4}$, C>G at the germline 0.2% still yields 2.0 expected carriers per route and a detection probability of 0.86 (§2.6), so C>G is supplied even at the reference rate and its magnitude cannot gate the hit list. The second reading is the one consistent with the benchmark in Table 2.

We therefore do not report a calibrated *in vitro* ENU C>G rate. Inverting the observation through the model-implied null gives 2.6%–26.2% (point estimate 9.7%), narrower and lower than the 7.0%–49.7% that the uncorrected inversion returns, but the inversion presumes that supply weighting sets the hit composition, which is the premise the preceding paragraph removes. The identifiable statement is the negative one: the germline spectrum is inconsistent with this output, and the screen behaves as though chemistry entered as a threshold rather than as a weight.

Table 3. Three C>G hits in thirteen industry ENU hits, tested against five nulls for the probability that a hit is C>G. The result depends entirely on whether the null carries the germline supply term, which is why the count identifies an inconsistency with that spectrum but not a corrected rate. Pooled $\sigma = \max_d \sigma_d$; $\beta = 1$.

Null	**Basis**	π(C>G)	$P(X \geq 3)$
N0 raw spectrum	supply only, pre-selection	0.0020	2.3×10^{-6}
N1 supply × composition	supply, class sizes	0.0020	2.4×10^{-6}
N3 model-implied	supply $\times \exp(\beta\sigma)$	0.0056	4.7×10^{-5}
N2 selection only (top-100)	σ ranking, no supply	0.24	0.64
N4 atlas composition	class sizes only	0.17	0.39

3.6 How far σ transports from the atlas to a Ba/F3 screen

Every recall figure in Table 2 assumes that σ, measured in H3122 cells at mild selection, reproduces the ordering a Ba/F3 screen would give at 5–15× IC_{50}. Because σ alone is the best-performing ranking in that table, this assumption carries the benchmark, and we test it two ways rather than asserting it: externally against measured Ba/F3 potency, and internally against the atlas's own dose series (Methods, §2.10).

External test. Transport is real but uneven, and it tracks how reproducible the underlying potency measurement is (Table 4). For zotizalkib the atlas ranks variants almost as measured potency does ($\rho = 0.82$, $n = 13$), and for alectinib the association is moderate ($\rho = 0.51$, $n = 18$). For lorlatinib there is no evidence of transport ($\rho = 0.17$, $n = 18$, bootstrap interval spanning zero). Two features of the lorlatinib data are worth recording, though we offer neither as an explanation. Its variants show the widest disagreement between laboratories of the three drugs, a median inter-study range of 0.32 $\log_{10}$ units, or 2.1-fold on the same variant; and the atlas assayed lorlatinib at 5 nM against a Ba/F3 wild-type IC_{50} of 1.35 nM, only 3.7×, whereas its zotizalkib arm sits at 30.6×. Dose ratio by itself does not order the three drugs, since alectinib's 1.7× is the narrowest of all and its transport is the intermediate one. At $n = 18$ we cannot

separate measurement noise from a genuine failure of transport, and we make no such claim. The zotizalkib estimate carries the opposite caveat: its variants are measured in a single study, so it has no cross-laboratory replication and is likely optimistic.

Internal test. The atlas also assays its exon-23 and exon-22 sub-libraries at several concentrations per drug, which asks the same question without leaving the assay. Across seven such series spanning 2- to 9-fold in dose, the rank correlation between a drug's lowest and highest concentration is 0.30 to 0.63 (median 0.39), and between 5 and 15 of the top 25 variants at the lowest dose remain in the top 25 at the highest. Adjacent steps agree better (median $\rho = 0.64$) than the full span, as expected if the effect accumulates with dose.

That accumulation is not all drug response. Two different drugs measured at matched dose rank agree at $\rho = 0.75$, better than a single drug agrees with itself across its own dose range, and matched dose ranks agree better than unmatched ones ($\rho = 0.75$ against 0.57), which points to batch structure shared across arms harvested together. Three further observations frame the result. The main atlas corresponds to the lowest concentration of each series, best-matching at 20 for alectinib and 5 for lorlatinib (zotizalkib matches its middle concentration instead, which we report without an explanation); dispersion rises monotonically with dose in every exon-23 and exon-22 series, so the condition the rest of this paper uses as σ is the least discriminating one available; and the main atlas agrees with its own nominal repeat in the sub-library at $\rho = 0.55$–0.62. The obvious confound does not apply: the sub-library window is not range-restricted but enriched, carrying 1.1 to 1.5 times the σ dispersion of the full atlas and 33 of its 100 highest-σ class-forced variants from 16% of the library.

Taken with the external test, σ carries real but limited rank information at the resolution available here. That bounds the benchmark in both directions. It is a further reason to read the chemistry null in Table 2 as a statement about resolution rather than about the absence of an effect, and a reason the reachability inventory, which depends on no ranking at all, is the part of this framework that does not inherit the limitation.

Proxy choice. The same data speak to the neladalkib proxy. The atlas does not contain neladalkib, so Table 2 and Figure 4 substitute another drug's σ. Against measured Ba/F3 neladalkib potency, lorlatinib and zotizalkib σ perform equally ($\rho = 0.57$ each, $n = 13$) and alectinib performs worse ($\rho = 0.21$). The zotizalkib proxy used in Figure 4 is therefore supported, a lorlatinib proxy would serve as well, and an alectinib proxy is the one choice the data do not support.

3.7 The MET screens observe the mutagen's output rather than infer it

For KRAS G12C and ALK the substitution class of a reported hit has to be inferred from the amino-acid change, because neither screen reports which base changed. The MET pipeline does report it. Tiedt et al. [12] give, for every resistant clone, the nucleotide alteration and the number of clones carrying it, numbered against RefSeq NM_000245.2, and they ran two screens with one

Table 4. Atlas σ against measured Ba/F3 cellular IC_{50} fold resistance for the same variants, Spearman rank correlation with a 10,000-sample bootstrap interval. Fold resistance is computed within each (study, drug) stratum against that study's own wild-type EML4-ALK v1 measurement. Spread is the median inter-study range in $\log_{10}$ units, zero where only one study measured the variant. Dose ratio is the atlas assay concentration over the Ba/F3 wild-type IC_{50}.

Drug	n	ρ	boot 95%	Spread	Dose ratio
Zotizalkib	13	0.82	0.49 to 0.96	0.00	30.6×
Alectinib	18	0.51	−0.04 to 0.86	0.13	1.7×
Lorlatinib	18	0.17	−0.43 to 0.72	0.32	3.7×

mutagen: a single 16-hour ENU exposure at 50 µg/mL on 3×10^7 Ba/F3 TPR-MET cells per arm, selected with either the type I inhibitor NVP-BVU972 or the type II inhibitor AMG 458. Estevam et al. [18] assayed both of those drugs in a chemistry-free MET kinase-domain deep mutational scan. The ENU class spectrum is therefore observed here, and the two screens form a control the other pipelines cannot supply.

One mutagen, two drugs. If the mutagen's class weighting set the composition of a hit list, two screens sharing a mutagen, a cell line, a protocol and a laboratory would produce the same class composition. They do not (Fig. 5A). The 85 NVP-BVU972 clones are 60% T>C and contain no C>G at all; the 60 AMG 458 clones are 38% T>C, 33% T>A and 15% T>G, with C>T almost absent. The difference between them is significant ($\chi^2 = 12.4$, df $= 5$, $P = 0.029$), and neither matches the germline ENU reference. Composition follows the drug.

One codon, every route. Under NVP-BVU972, 75 of 85 clones carry a substitution at Y1230, whose wild-type codon is TAT. That codon has exactly six single-nucleotide missense routes, and the screen recovered five of them: Y1230C (31 clones), Y1230H (20), Y1230N (19), Y1230S (4) and Y1230D (1) (Fig. 5B). Those five span three substitution classes whose germline ENU supply differs 7.3-fold, and the screen recovered all five regardless. Within the codon the clone counts do track supply: the two T>G routes predict 5.1 clones between them and delivered 5. The hit list is therefore supply-independent while the frequencies inside it are supply-dependent, which is the threshold-not-weight claim at nucleotide resolution.

The sixth route, Y1230F, was not recovered, and we cannot account for it. Its class is T>A, which the reference places at 31%; its atlas rank is 54 of 5,458, above four variants the screen did find; and its DMSO-arm score is higher than all four of those, so it is not a growth defect. Tiedt et al. noted the same absence. We report it as an unexplained discrepancy rather than fit an account to it.

Recovery and the two-step blind spot. Every reported hit is present in the atlas, 10 of 10 for NVP-BVU972 and 13 of 13 for AMG 458, at median ranks of 171 and 229 of 5,458. Against that, three-quarters of each drug's chemistry-free top 100 is out of one-step reach: 72 of 100 under NVP-BVU972 and 75 of 100 under AMG 458 require two or more nucleotide changes (Fig. 5C). This is a far larger structural blind spot than the ALK pipeline can display, and the reason is

library design rather than biology. The Oh prime-editing atlas contains only variants reachable in one nucleotide step, so it cannot measure this quantity at all; the MET scan and the KRAS Twist library both include multi-nucleotide variants and can (§2.11). Results are reported on the exon-14-deleted background, which is the one in which MET inhibitor resistance arises clinically and which recovers the known resistance variants best. The wild-type background is weaker on both counts: it holds 7 of the 10 NVP-BVU972 hits and 11 of the 13 AMG 458 hits, at median ranks of 495 and 2,472.

4 Discussion

Mutagen bias in preclinical TKI-resistance screens has been acknowledged as a limitation of individual screens for at least a decade [2, 5], and recent screens continue to flag it in prose while reporting unadjusted top-N hit lists [7, 14]. The present work complements rather than replaces these screens. Anchoring $\sigma(v)$ in three chemistry-free atlases lets us enumerate, before a screen is run, which reported resistance variants the chosen chemistry cannot produce in a single nucleotide step, patient-observed and *in vitro* ones labelled separately, and bounds how far the ranking of the remaining variants departs from the chemistry-free one. The field is arriving at the same reachability filter one case at a time. Wang et al. [17], in their EGFR L858R DMS against BLU-945, restrict candidate resistance mutations to Hamming distance 1 from the wild-type codon and conclude that M793 escapes are unlikely to be observed clinically because they require multi-nucleotide changes. We apply that filter systematically to every (variant, mutagen, drug) combination in the preclinical literature.

The reachability category is the sharper of the two, and it has two tiers. The hard tier is exact: four of six substitution classes carry a zero rate under a base-editor pair, so no depth recovers a variant whose only route lies in one of them. The soft tier is the two-step one. R68L and V8M each require two nucleotide changes from the wild-type codon, and R68L is the top-ranked variant in the chemistry-free atlas under both drugs; both are suppressed by roughly two orders of magnitude at practical depth rather than forbidden (§2.9), which is consistent with error-prone PCR compound screens recovering multi-nucleotide events at all. Both variants come from the accompanying *in vitro* deep mutational scan [9] and not from that study's clinical cohort, so the claim attached to them is codon-level reachability and not clinical recurrence.

Base editing is a methodological advance over chemical mutagenesis, and it carries a chemistry bias of its own: a cytosine and an adenine base editor together restrict the library to two of six substitution classes. That restriction excludes 7 of the 12 KRAS G12C panel variants outright, among them the patient-observed H95D and Y96D (Figure 3); two more, V8M and R68L, lie two nucleotide changes from the wild-type codon and are out of reach for any single-nucleotide chemistry, leaving 3 of the 12 that a base-editor pair can produce. A published BE-based KRAS G12C resistance screen [15] cannot discover these variants, not because its drug-selection measurement is wrong but because its library never contains them. Where several clinical variants fall in the four BE-unreachable classes, a BE screen needs at least one chemistry-free construction

step to be clinically comprehensive.

The quantitative category depends on the mutagen's spectrum and the variant's forced class, and the framework yields a per-variant prediction for every (gene, mutagen, drug) combination. Patient-observed H95D (class C>G) is under-counted by +47 ranks under Mutazyme II (§3.3), and L1152R (class T>G) carries the largest predicted shift in the ALK panel, +597 ranks at $\beta = 1$ under alectinib. Three concrete predictions follow for the current pipelines. An ENU-based neladalkib screen [14] will under-count variants whose forced class is C>G or T>G. ENU-based KRAS G12C screens will under-count Y96D (T>G) and the patient-observed H95D (C>G), the latter by the margin its 0.2% class rate sets rather than by exclusion, since that rate still supplies the variant at practical depth (§2.6). Base-editing KRAS G12C screens will miss C12F, V8L, Y96D, H95D, H95Q, R68S, and R68M by construction. Each prediction is enumerable from Supplementary Table S1 and checkable against a screen's reported hit list.

Two compound-mutation settings are directly checkable in the same way. The error-prone PCR neladalkib screen on an I1171N background [7] recovers five compound partners (L1256F, E1129V, E1129K, C1156Y, T1151R), and the chemistry side of the decomposition identifies which of their forced classes Mutazyme II supplies well (Supplementary Table S1). Clinical characterization of NVL-655 in ALKOVE-1 [26] reports responses in patients carrying cis-allelic $\mathrm{ALK}^{\mathrm{G1202R/F1174C}}$ and co-occurring $\mathrm{ALK}^{\mathrm{G1202R}}$ with $\mathrm{ALK}^{\mathrm{G1269A}}$, alongside reduced activity against the $\mathrm{ALK}^{\mathrm{I1171N/D1203N}}$ compound. The supply partition of these partners follows from the same codon and spectrum inputs.

Six limitations bound the framework. The forced-class classification depends on the exact wild-type codon, and the KRAS-4A NM_033360.4 table used here is ClinVar-anchored at every resistance-panel residue but consensus-only for 39 residues in the 4A-specific hypervariable region (positions 151–189). The KRAS G12C panel is composite, so the structural finding for R68L and V8M is a codon-level claim that holds regardless of clinical recurrence; assessing recurrence would require patient-cohort data we do not have for these two variants. The KRAS drug-selection score is a single-replicate normalized Z and is noisier than the Oh Resistance score, which accounts for the lower top-K recovery in KRAS than in ALK. The ENU spectrum is the weakest input here: no pre-selection six-class spectrum measured on an *in vitro* ENU-treated mammalian library exists, the mouse germline compilation we substitute is phenotype-ascertained, and §3.5 shows it is inconsistent with the one *in vitro* ENU output we can observe without identifying what should replace it. The model is "supply × selection" at one substitution step, so it does not predict variants that arise through compound selection, multi-step mutagenesis protocols, or sgRNA-independent secondary edits, and a screen designed to catch compound mutations needs the conditional second-step supply term given in Methods. The anchoring atlases measure drug response at mild selection (20 nM alectinib, 5 nM lorlatinib, 300 nM zotizalkib [11]) whereas Ba/F3 screens typically run at 5–15× IC_{50}, so a variant with a small IC_{50} shift may have $\sigma > 0$ at mild drug and $\sigma \approx 0$ at strong drug. Section 3.6 tests this, and the answer is drug-dependent: against measured Ba/F3 potency, transport is strong for zotizalkib, moderate for alectinib and absent for lorlatinib, and within the atlas's own dose series the rank order moves substantially

across a 2- to 9-fold change in concentration. This is the sharpest limitation in the paper. It does not favor either ranking in Table 2, since both are built on the same σ, but it does mean the comparison is made at low resolution, and it is a further reason the reachability inventory rather than the ranking is what we put forward. None of these limitations change the sign or direction of the rank-shift and recovery predictions. They bound the strength of individual point estimates.

The uniform-1/6 chemistry-free baseline is the right reference for *fitness under drug*, because it separates what the atlas measures from what a mutagen supplies. It is *not* the reference for what a tumour can acquire somatically. Smoking-associated NSCLC, the context of the KRAS G12C cohort, is dominated by COSMIC SBS4, a tobacco-adduct signature concentrated in C>A substitutions, whereas ALK-rearranged LUAD in never-smokers is dominated by the clock-like SBS1 and SBS5 [27]. Neither matches a preclinical chemistry one-for-one: Mutazyme II's dominant C>T is reasonably concordant with SBS1 and SBS5, ENU's dominant T>C and T>A are emphasized by neither SBS4 nor SBS1, and the CBE+ABE pair covers C>T and T>C rather than SBS4's C>A. The mismatch sharpens the structural claim, because the most informative blind spot is a variant that is tumour-reachable under the relevant signature mixture and mutagen-unreachable under the chosen chemistry. Supplementary Table S1 tags each panel variant's forced class with the published signatures that emphasize it. We defer the quantitative, trinucleotide-context-accurate analysis to a companion study, which requires per-patient 96-channel catalogs from GENIE for both cohorts.

The proposal itself is simple. Anchor $\sigma(v)$ in a chemistry-free atlas, enumerate what the chosen mutagen cannot supply, and the resulting predictions become testable before the screen runs and auditable against its reported hit list. MET is the third pipeline we audit here (§3.7), and it is the one that settles the mechanism, because its screen reports nucleotides and its two arms share a mutagen. The same test is available for any pipeline with those two halves. For EGFR, the L858R DMS [17] provides the chemistry-free side and several published ENU- or ePCR-based screens the chemistry-biased side. Each new chemistry-free atlas brings a further TKI pipeline inside the framework, whichever construction method produced it. Until such atlases cover every pipeline, chemistry-biased screens will remain the field's first line of resistance prediction, and the variants each one will systematically miss are already enumerable from codon tables and mutagen spectra alone.

Availability of data and materials

All codon tables, substitution spectra, derived tables, figure scripts, and analysis code are released under a permissive license at https://github.com/allkure/mutagen-reachability, archived at `DOI_TO_BE_INSERTED`. Every quantitative claim in this manuscript is produced by a named script and recorded in the master register `paper/NUMBERS.md`, which gives for each value its source file, the function that computed it, and the place it appears here; two audit scripts check that register against the manuscript in both directions.

The primary data are redistributed only as derived long-format tables with explicit back-references to their sources: the chemistry-free saturation mutagenesis atlases of KRAS G12C [10] and ALK [11], the MET kinase-domain deep mutational scan [18], the two ENU MET resistance screens whose hit lists and clone counts are transcribed from Supplementary Tables S1 and S2 of [12], and the germline ENU [16] and Mutazyme II [2] substitution spectra. All are publicly available through their original publications.

The per-residue reachability inventory behind Fig. 3C is provided as an interactive track covering all three kinases, every mutagen chemistry and every drug arm, together with print-resolution vector versions of each gene-and-drug combination.

Competing interests

The author declares no competing interests.

Author contributions

MR designed the study, performed the analysis, and wrote the manuscript.

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Figures

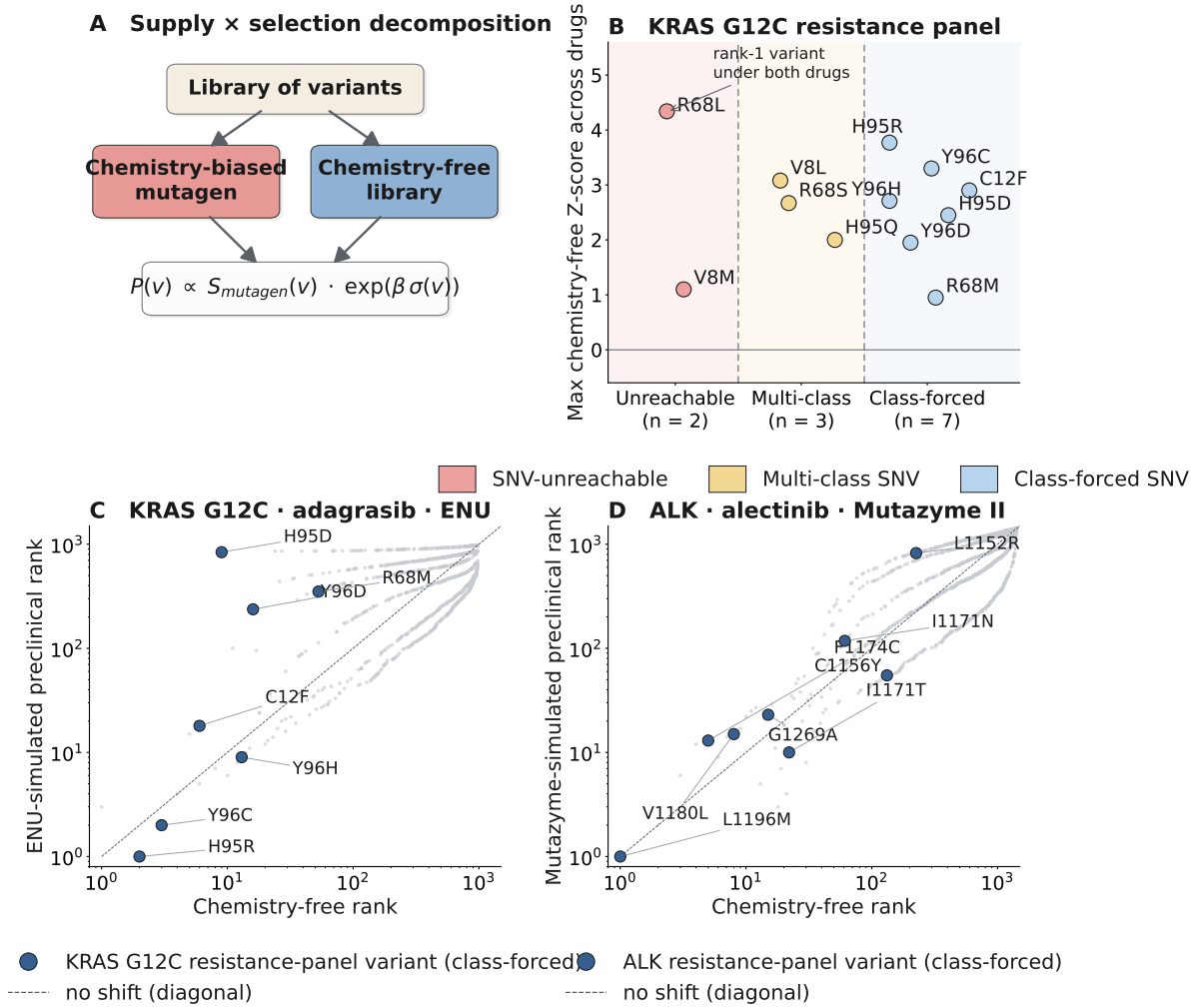


Figure 1. Chemistry-biased mutagens reorder and sometimes forbid preclinical TKI-resistance rankings. **A**, The supply × selection decomposition, $P(v) \propto S_{\text{mutagen}}(v) \cdot \exp(\beta\sigma(v))$. **B**, The 12 composite KRAS G12C panel variants (Table S1), binned by single-nucleotide reachability from the wild-type codon; y, maximum chemistry-free Z-score across sotorasib and adagrasib. **C**, **D**, Simulated against chemistry-free rank, both axes log scale, for KRAS G12C under ENU with adagrasib (1,205 variants [10]) and for ALK under Mutazyme II with alectinib (1,427 variants [11]); dashed line, no shift.

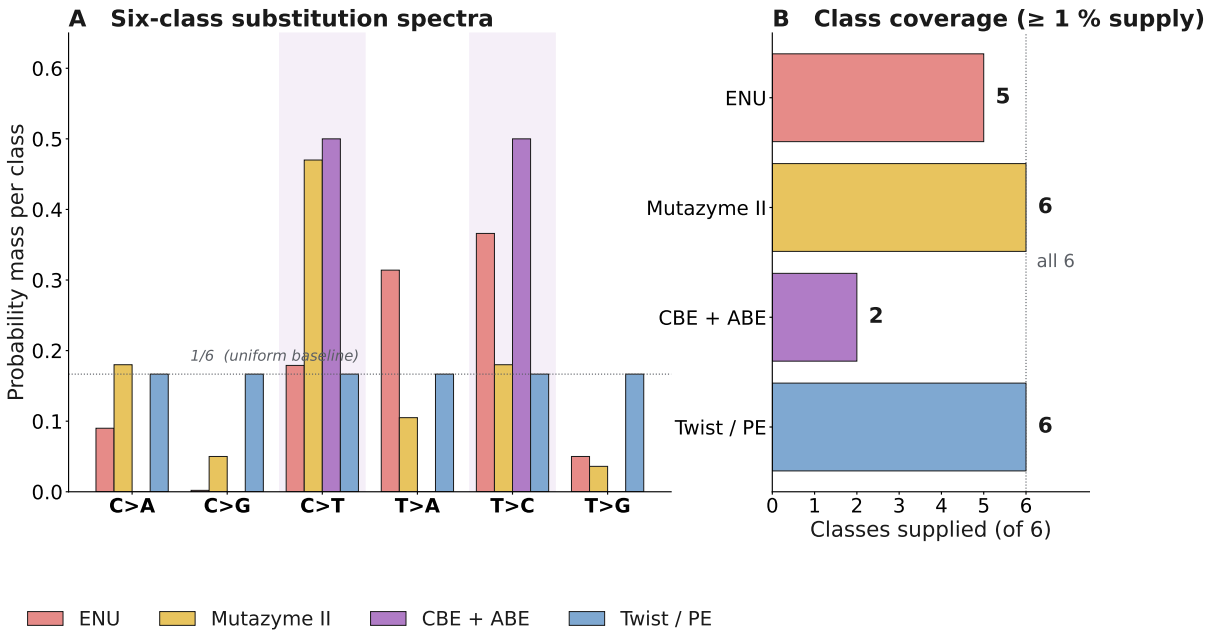


Figure 2. Mutagen chemistries used in current preclinical TKI-resistance screens. **A**, Six-class pyrimidine substitution spectra of ENU [16], Mutazyme II [2], and combined CBE+ABE, against the chemistry-free uniform baseline (1/6 per class, dotted). The lavender band marks the two base-editing-reachable classes. **B**, Number of classes each mutagen supplies at $\geq 1\%$.

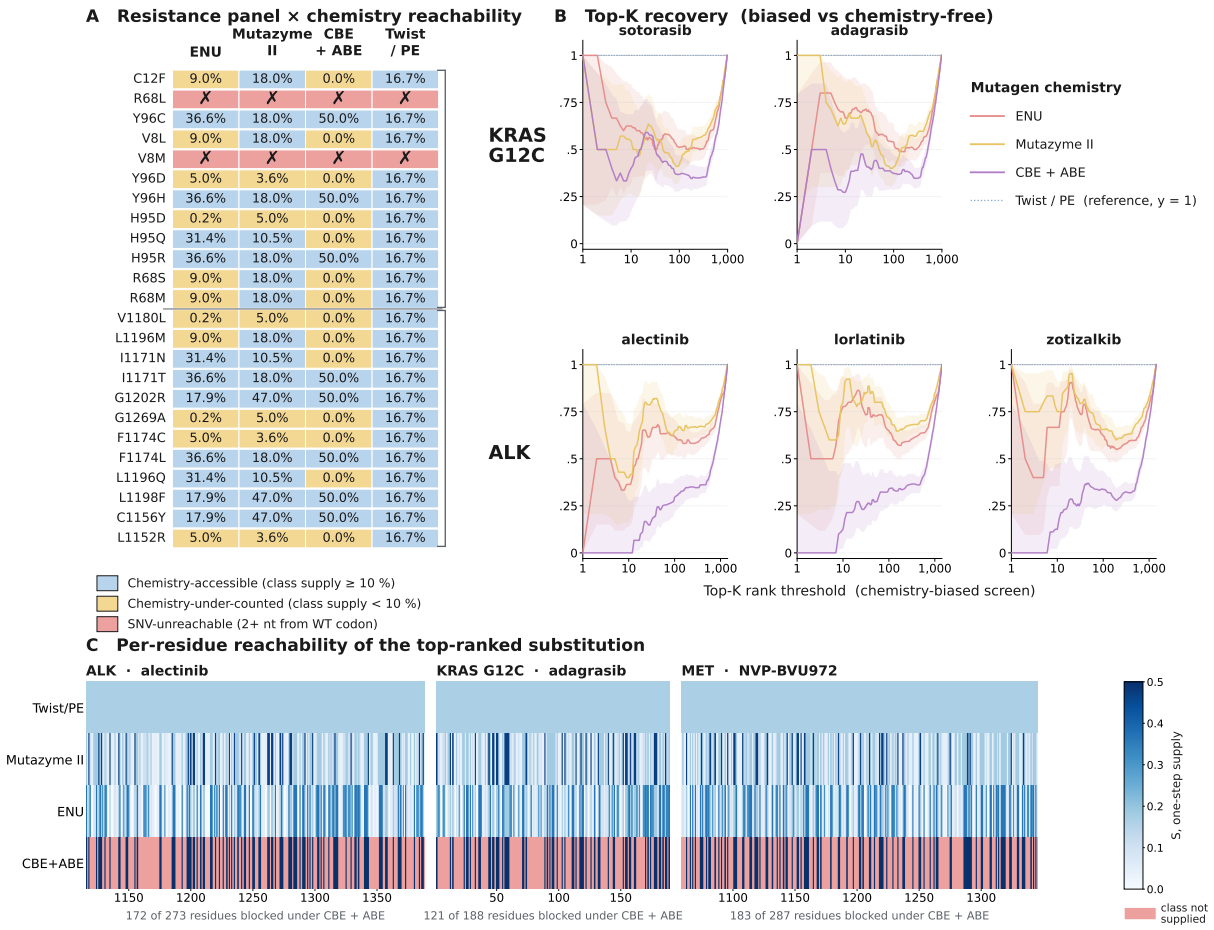


Figure 3. Chemistry blind-spots and recovery of priority variants from the chemistry-free atlases. **A**, Reachability of the 24 panel variants (KRAS G12C above the divider, ALK below; Table S1) under the four chemistries of Table 1. Blue, accessible (maximum class supply $\geq 10\%$); amber, under-counted ($< 10\%$); red, two-step from the wild-type codon. Labels give the maximum class supply. **B**, Fraction of the chemistry-free top-K also present in the chemistry-biased top-K, one subpanel per gene and drug, with 95% Wilson bands and K on a log scale. The chemistry-free reference is 1 by construction (dotted). Colors as in Fig. 2A. **C**, For every residue of each kinase domain, the one-step supply each chemistry gives the substitution the chemistry-free atlas ranks highest at that position (ALK with alectinib, KRAS G12C with adagrasib, MET with NVP-BVU972; the other arms are in the interactive inventory). The view covers class-forced substitutions only, so a chemistry counts as blocked here just where its spectrum gives that class zero; variants needing two or more nucleotide changes lie outside it by construction and are reported in §3.2 and §3.7. On that scope ENU and Mutazyme II block no residue in any of the three genes, while CBE+ABE blocks 63%–66% of them.

Retrospective validation of $P(v) = S_{\text{mutagen}}(v) \cdot \sigma(v)$ across three preclinical ALK TKI resistance screens

A Mutazyme II × alectinib (single-mutation anchor)

Variant	WT codon	Reachability	Class	Rate (%)	Chem-free rank	Simulated rank
V1180L	GTT	class-forced	C>G	5.0	8	15
L1196M	CTG	class-forced	C>A	18.0	1	1
I1171N	ATC	class-forced	T>A	10.5	61	118
I1171T	ATC	class-forced	T>C	18.0	22	10
L1198F	CTC	class-forced	C>T	47.0	18	3
G1202R	GGA	multi-class	—	—	—	—

B ePCR × neladalkib (compound-mutation, I1171N fixed-V1)

Variant	WT codon	Reachability	Class	Rate (%)	Chem-free rank	Simulated rank
L1256F	CTC	class-forced	C>T	47.0	124	44
E1129V	GAG	class-forced	T>A	10.5	642	726
E1129K	GAG	class-forced	C>T	47.0	854	266
C1156Y	TGC	class-forced	C>T	47.0	463	125
T1151R	ACG	class-forced	C>G	5.0	567	985

C ENU × neladalkib + alectinib + lorlatinib (industry screen, aggregate)

Variant	WT codon	Reachability	Class	Rate (%)	Chem-free rank	Simulated rank
I1171N	ATC	class-forced	T>A	31.4	61	40
I1171S	ATC	class-forced	T>G	5.0	26	34
I1171T	ATC	class-forced	T>C	36.6	22	6
F1174C	TTC	class-forced	T>G	5.0	5	10
F1174V	TTC	class-forced	T>G	5.0	16	20
V1180L	GTT	class-forced	C>G	**0.20**	8	229
T1151K	ACG	class-forced	C>A	9.0	72	165
T1151R	ACG	class-forced	C>G	**0.20**	104	1190
E1129K	GAG	class-forced	C>T	17.9	689	561
R1192P	?	unanchored	—	—	—	—
L1196M	CTG	class-forced	C>A	9.0	1	2
G1269A	GGA	class-forced	C>G	**0.20**	15	401
G1202R	GGA	multi-class	—	—	—	—

Figure 4. Retrospective validation of $P(v) \propto S_{\text{mutagen}}(v) \cdot \exp(\beta\sigma(v))$ across three preclinical ALK TKI resistance screens. Each panel lists the screen's observed variants with wild-type codon, reachability, forced substitution class, class supply under the screen's mutagen, and chemistry-free against simulated rank. **A**, Mutazyme II with alectinib [2]; six variants, five class-forced (G1202R is multi-class and omitted). **B**, ePCR neladalkib compound screen on an I1171N background [7]; five V2 partners. **C**, Industry ENU screen across alectinib, lorlatinib, and neladalkib [14]; thirteen aggregate hits, R1192P outside the atlas residue window. Panels B and C use zotizalkib σ as the fourth-generation proxy.

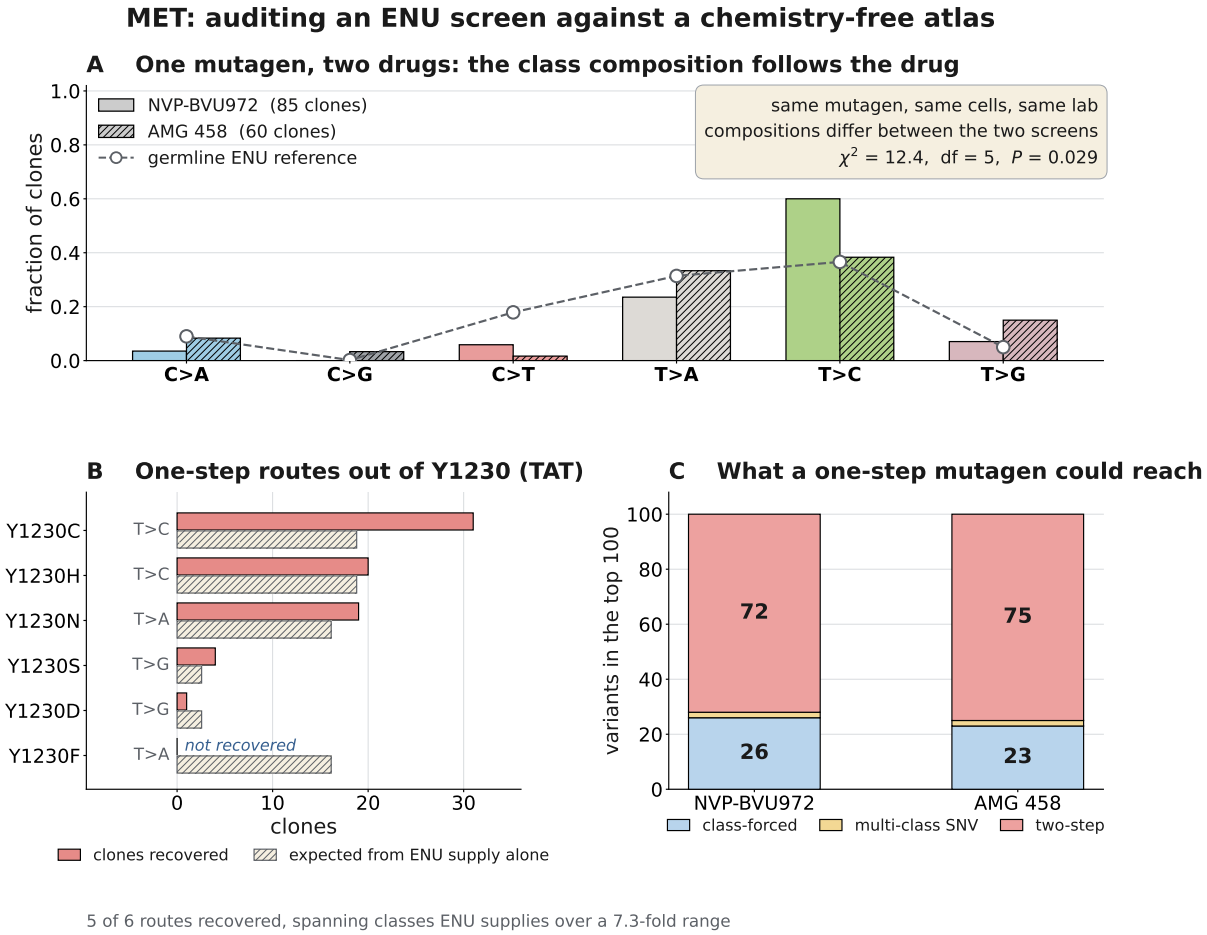


Figure 5. The MET audit. **A**, Six-class composition of the clones each Tiedt screen recovered, taken from the nucleotide alterations that study reports, against the germline ENU reference (dashed). One mutagen, one cell line, one laboratory, two drugs. **B**, Every single-nucleotide missense route out of the Y1230 wild-type codon TAT, with its substitution class, the clones recovered, and what the ENU class supply alone predicts. **C**, Reachability of each drug's chemistry-free top 100 from the wild-type codon.

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